Monte Carlo Estimation of π and Gaussian Integrals

1 Introduction

This report explores Monte Carlo methods for numerical integration. We begin with a geometric approach to estimating π, and then extend the idea to compute integrals involving Gaussian functions. Finally, we introduce importance sampling as a variance reduction technique.

2 Estimating π Using Monte Carlo

Consider a unit circle inscribed inside a square of side length 2 centered at the origin.

The equation of the circle is:

$$x^2 + y^2 \leq 1$$

The area of the circle is π, and the area of the square is 4. Thus, the ratio of the areas is:

$$\frac{\text{Area of circle}}{\text{Area of square}} = \frac{\pi}{4}$$

We can estimate this ratio using random sampling.

2.1 Method

We generate random points (x, y) uniformly in the square $[-1, 1] \times [-1, 1]$ and count how many fall inside the circle. The number of points falling insdie the circle is directly proportional to the area of the circle Then,

$$\pi \approx 4 \cdot \frac{\text{Number of points inside circle}}{\text{Total number of points}}$$

2.2 Simulation Result

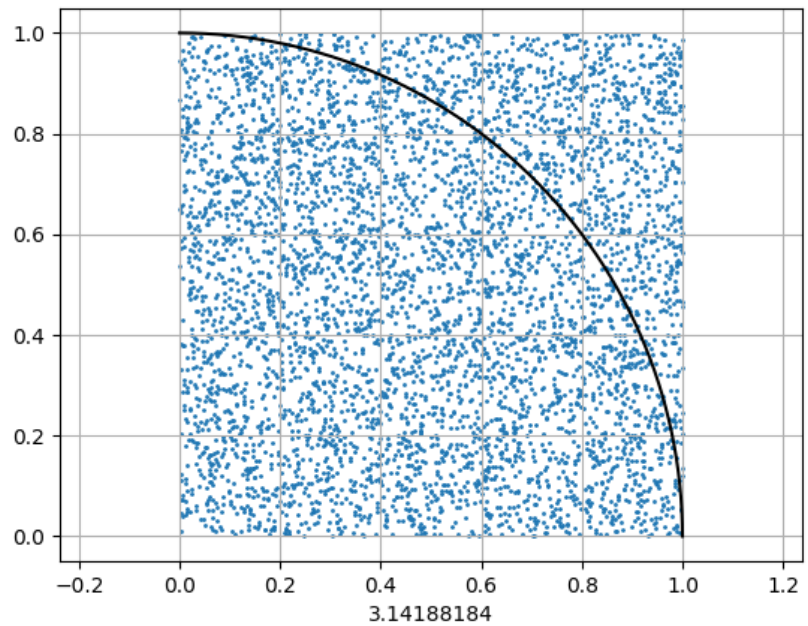

Figure 1: Estimation of π

2.3 Limitations

The Monte Carlo estimation of π using uniform sampling has a limitation from statistical variance. The estimator is based on counting random points that fall inside the unit circle, which introduces randomness in the final estimate.

Let X_i be an indicator random variable defined as:

$$X_i = \begin{cases} 1 & \text{if the point lies inside the circle} \\ 0 & \text{otherwise} \end{cases}$$

Then the estimator for π can be written as:

$$\hat{\pi} = 4 \cdot \frac{1}{N} \sum_{i=1}^{N} X_i$$

The expectation of each X_i is:

$$\mathbb{E}[X_i] = \frac{\pi}{4}$$

Since X_i is a Bernoulli random variable, its variance is:

$$\mathrm{Var}(X_i) = \frac{\pi}{4}\left(1 - \frac{\pi}{4}\right)$$

The variance of the estimator is:

$$\mathrm{Var}(\hat{\pi}) = 16 \cdot \frac{1}{N} \mathrm{Var}(X_i)$$

$$\mathrm{Var}(\hat{\pi}) = \frac{16}{N} \cdot \frac{\pi}{4}\left(1 - \frac{\pi}{4}\right)$$

This shows that the standard deviation of the estimator decreases as:

$$\mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$$

Thus, to significantly improve accuracy, the number of samples N must be increased quadratically.

This motivates the need for variance reduction techniques such as importance sampling in more general Monte Carlo integration problems.

3 Gaussian Integral

We now consider the Gaussian integral:

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx$$

The exact value is known to be:

$$I = \sqrt{\pi}$$

3.1 Truncated Uniform Sampling

We approximate the integral over a finite interval $[0, 10]$ as the value becomes sufficiently small by that x:

$$I \approx 2 \int_{0}^{10} e^{-x^2} dx$$

Using Monte Carlo:

$$I \approx 20 \cdot \frac{\text{Number of points under curve}}{\text{Total points}}$$

4 Importance Sampling

Uniform sampling is inefficient because many samples fall in regions where $f(x)$ is nearly zero.

Importance sampling improves efficiency by sampling from a distribution similar to the integrand.

4.1 General Idea

We rewrite the integral as:

$$\int f(x)\,dx = \int \frac{f(x)}{p(x)}\,p(x)\,dx$$

Now this matches the definition of expectation with respect to a probability density $p(x)$:

$$\mathbb{E}_p[g(x)] = \int g(x)\,p(x)\,dx$$

Substituting $g(x) = \frac{f(x)}{p(x)}$, we obtain:

$$I = \mathbb{E}_p\left[\frac{f(x)}{p(x)}\right]$$

This expression highlights the role of $p(x)$ as a weighting function. In the expectation, $p(x)$ determines how frequently different values of x are sampled, effectively assigning higher probability (and thus more weight) to regions where $p(x)$ is large.

In this sense, $p(x)$ acts as the distribution that governs the random sampling same way p(x) governs Expectation of a distribution, while simultaneously influencing the contribution of each sample through the weighting process.

4.2 Choice of Distribution

For the Gaussian integrand $f(x) = e^{-x^2}$, we choose $p(x)$ such that it resembles the shape of $f(x)$ and reduces variance.

The idea is to sample more frequently in regions where $f(x)$ is large. This ensures that the most significant contributions to the integral are well represented.

An appropriate choice is:

$$p(x) = \mathcal{N}(0,1) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$$

This distribution closely matches the decay behavior of $f(x)$ and therefore improves sampling efficiency.

p(x) acts as the weights for our random number distribution.

4.3 Estimator

The Monte Carlo estimator becomes:

$$I \approx \frac{1}{N}\sum_{i=1}^{N}\frac{f(x_i)}{p(x_i)}$$

Substituting $f(x) = e^{-x^2}$ and $p(x)$:

$$I \approx \frac{1}{N}\sum_{i=1}^{N}\sqrt{2\pi}\,e^{-x_i^2/2}$$

where $x_i \sim \mathcal{N}(0,1)$.

4.4 Simulation Result

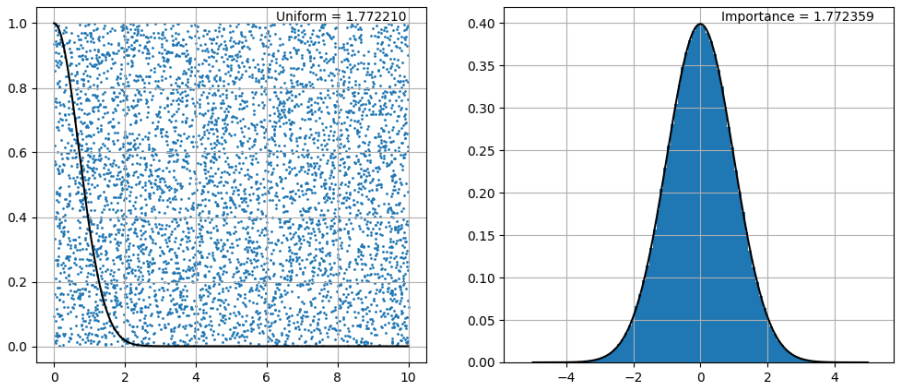

Figure 2: Comparison between OG Monte Carlo and Importance Sampling

5 Comparison

- Uniform sampling spreads points evenly but wastes samples in low-contribution regions.
- Importance sampling concentrates samples where the function is large.
- This reduces variance and improves convergence.